Exercise: Assume that X and Y are independent.

Show that $E(h_1(X)h_2(Y)) = E(h_1(X))E(h_2(Y))$.

$$E(h_1(X)h_2(Y)) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h_1(x)h_2(y) f_{X,Y}(x,y)\,dx\,dy$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h_1(x)h_2(y) f_X(x) f_Y(y)\,dx\,dy \quad \text{since } X, Y \text{ ind.}$$

$$= \left[\int_{-\infty}^{\infty} h_1(x) f_X(x)\,dx\right]\left[\int_{-\infty}^{\infty} h_2(y) f_Y(y)\,dy\right]$$

$$= E(h_1(X)) \qquad E(h_2(Y))$$

Comment: $E(XY) = E(X)E(Y)$ means X, Y are uncorrelated

Exercise: Assume that U_1 and U_2 are independent uniform$(0, 1)$ random variables. Let $U_{(1)}$ denote the minimum of U_1 and U_2, and let $U_{(2)}$ denote the larger of those two. Finally, let

$$L_1 = U_{(1)}, \quad L_2 = U_{(2)} - U_{(1)}, \quad \text{and} \quad L_3 = 1 - U_{(2)}.$$

Note that this is the "stick-broken-into-three-pieces" example, with L_i denoting the lengths of the three pieces ordered from left to right.

1. What is the joint distribution of (U_1, U_2)?
2. What is the joint distribution of $(U_{(1)}, U_{(2)})$?
3. What is the joint distribution of (L_1, L_2)?
4. Show that L_1, L_2, and L_3 each have the Beta$(1, 2)$ distribution. (Do not take it as "obvious" that all three have the same distribution.)

Comment: It is not difficult to see that if k uniform cuts are made, the length of each piece has the Beta$(1, k)$ distribution.

1 Joint distribution for (U_1, U_2)

$$f_{U_1,U_2}(u_1, u_2) = f_{U_1}(u_1)\, f_{U_2}(u_2)$$

since U_1, U_2 independent

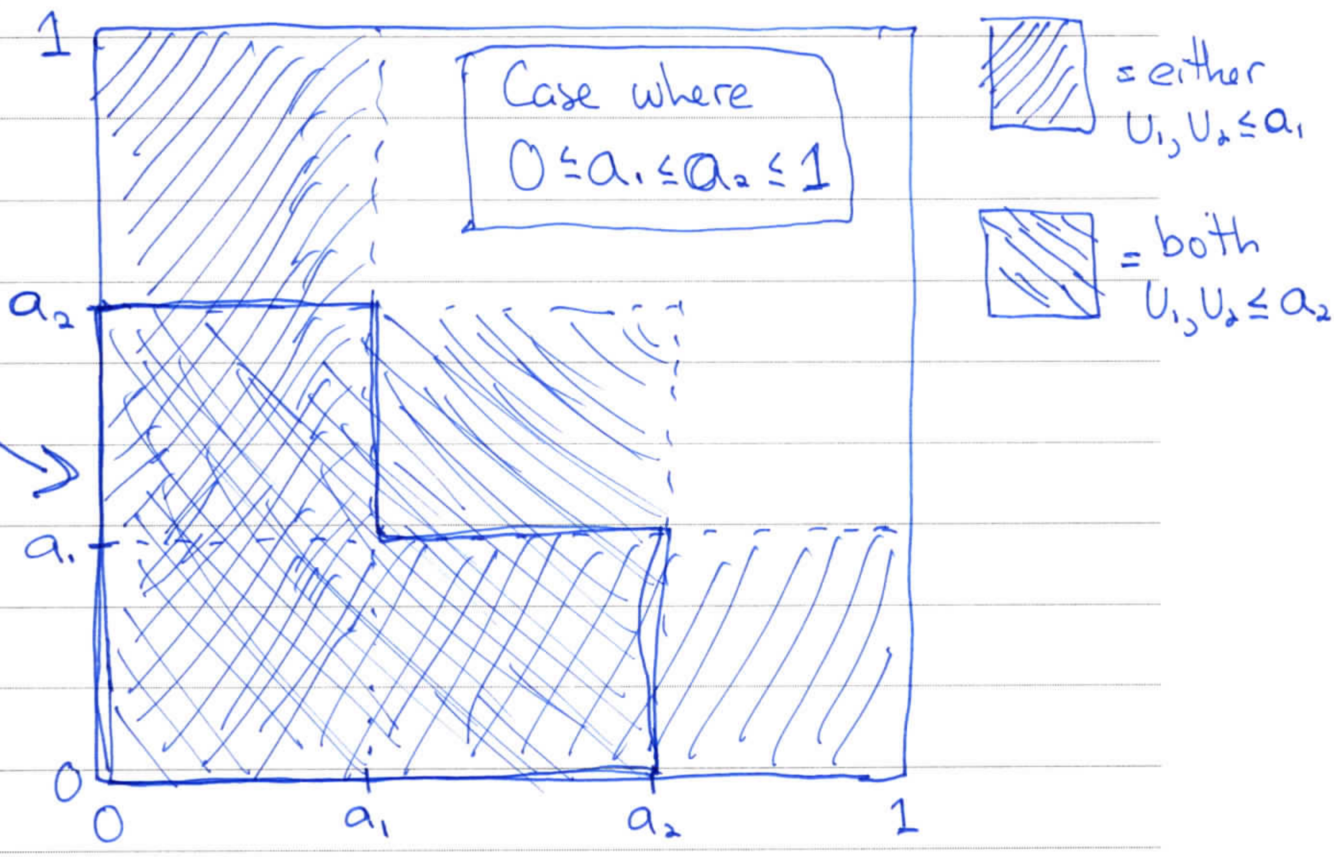
$$= \begin{cases} 1, & 0 \le u_1 \le 1,\ 0 \le u_2 \le 1 \\ 0, & \end{cases}$$

② Joint distribution for $(U_{(1)}, U_{(2)})$.

Find the CDF for $(U_{(1)}, U_{(2)})$

$$F_{U_{(1)}, U_{(2)}}(a_1, a_2) = P(U_{(1)} \le a_1, U_{(2)} \le a_2)$$

When does it hold that $U_{(1)} \le a_1$ and $U_{(2)} \le a_2$?
Need at least one of U_1, U_2 to be less than a_1, and both U_1, U_2 to be less than a_2.



We need to consider how the CDF varies as we change a_1, a_2

There is no requirement that $a_1 \le a_2$.

If $0 \le a_1 \le a_2 \le 1$, then $F_{U_{(1)},U_{(2)}}(a_1, a_2) =$

$$a_1^2 + 2(a_2 - a_1)a_1$$
$$= 2a_1a_2 - a_1^2$$

If $0 \le a_2 \le a_1 \le 1$ or $0 \le a_2 \le 1$ and $a_1 > 1$,

then $F_{U_{(1)},U_{(2)}}(a_1, a_2) = a_2^2$

If $a_1 < 0$ or $a_2 < 0$, then

$$F_{U_{(1)},U_{(2)}}(a_1, a_2) = 0$$

If $0 \le a_1 \le 1$, $a_2 > 1$, then

$$F_{U_{(1)},U_{(2)}}(a_1, a_2) = P(\text{at least one of } U_1, U_2 \le a_1)$$
$$= 1 - (1 - a_1)^2$$
$$= 2a_1 - a_1^2$$

This completely specifies the CDF

Now, take derivative to get density for $(U_{(1)}, U_{(2)})$

$$f_{U_{(1)},U_{(2)}}(a_1,a_2) = \frac{\partial F_{U_{(1)},U_{(2)}}(a_1,a_2)}{\partial a_1 \, \partial a_2}$$

$$= \begin{cases} 2, & 0 \le a_1 \le a_2 \le 1 \\ 0, & \text{otherwise} \end{cases}$$

Hence $(U_{(1)}, U_{(2)})$ is uniform on the triangle $0 \le a_1 \le a_2 \le 1$

(3) On HW

The Conditional Distributions

If we know (condition on) $Y = y$, what is the distribution of X? Such a question is answered by considering the **conditional distribution** of X given Y.

In the **discrete case,**

$$= \frac{P(X=x, Y=y)}{P(Y=y)}$$

$$f_{X|Y}(x|y) = P(X = x|Y = y) = \begin{cases} f_{X,Y}(x,y) \Big/ f_Y(y), & f_Y(y) > 0 \\ 0, & f_Y(y) = 0 \end{cases}$$

and in the **continuous case** we have

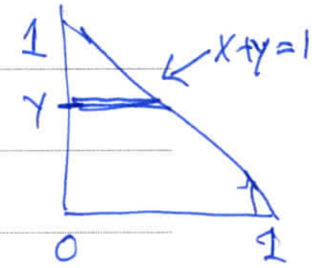
$$f_{X|Y}(x|y) = \begin{cases} f_{X,Y}(x,y) \Big/ f_Y(y), & f_Y(y) > 0 \\ 0, & f_Y(y) = 0 \end{cases},$$

with $f_{X|Y}$ called the **conditional density of X given Y.**

Comment: The continuous case is notable in that the event being conditioned on $(Y = y)$ has probability zero. There is nothing mathematically or conceptually incorrect about conditioning on probability zero events.

Exercise: Derive the conditional distribution of X given Y in our "triangular support" example.

$$f_{X,Y}(x,y) = 24xy, \quad x>0, \; y>0, \; x+y<1$$



$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

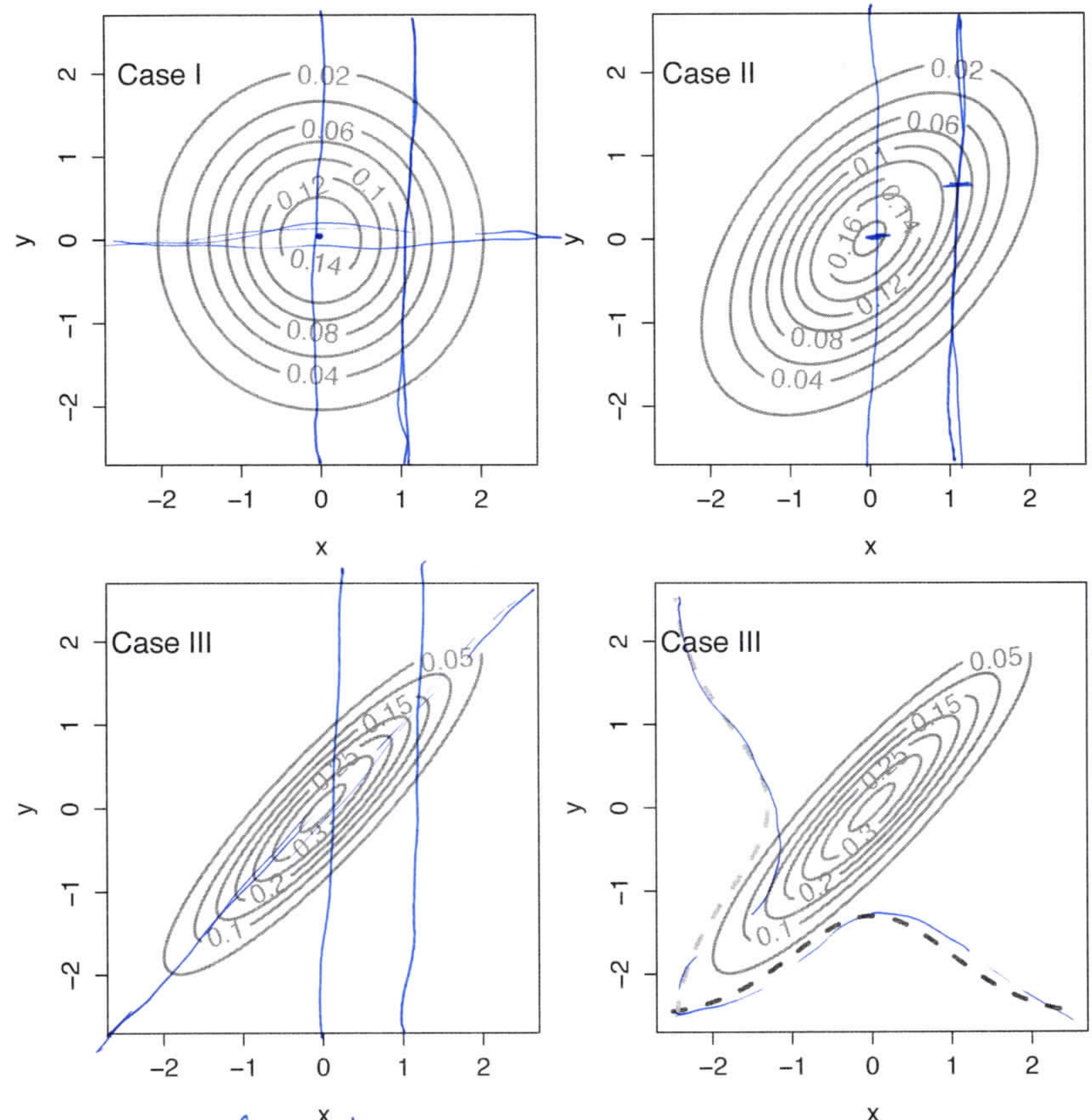
$$= \begin{cases} \dfrac{24xy}{12y(1-y)^2}, & 0<y<1, \; 0<x<1-y \\ 0, & \text{otherwise} \end{cases}$$

Note that when condition on $Y=y$, the support for X is $(0, 1-y)$

Example: This example illustates the distinction between the marginal and conditional distributions. Here, the bivariate pdf is depicted using **contour plots**.

In **Case I** shown below, X and Y are independent, and every conditional distribution is the same. In **Case II** there is a moderate amount of dependence. In **Case III** the dependence is stronger.

The fourth plot again shows Case III, but with the marginals imposed. In fact, the marginals are the same in all three Cases.



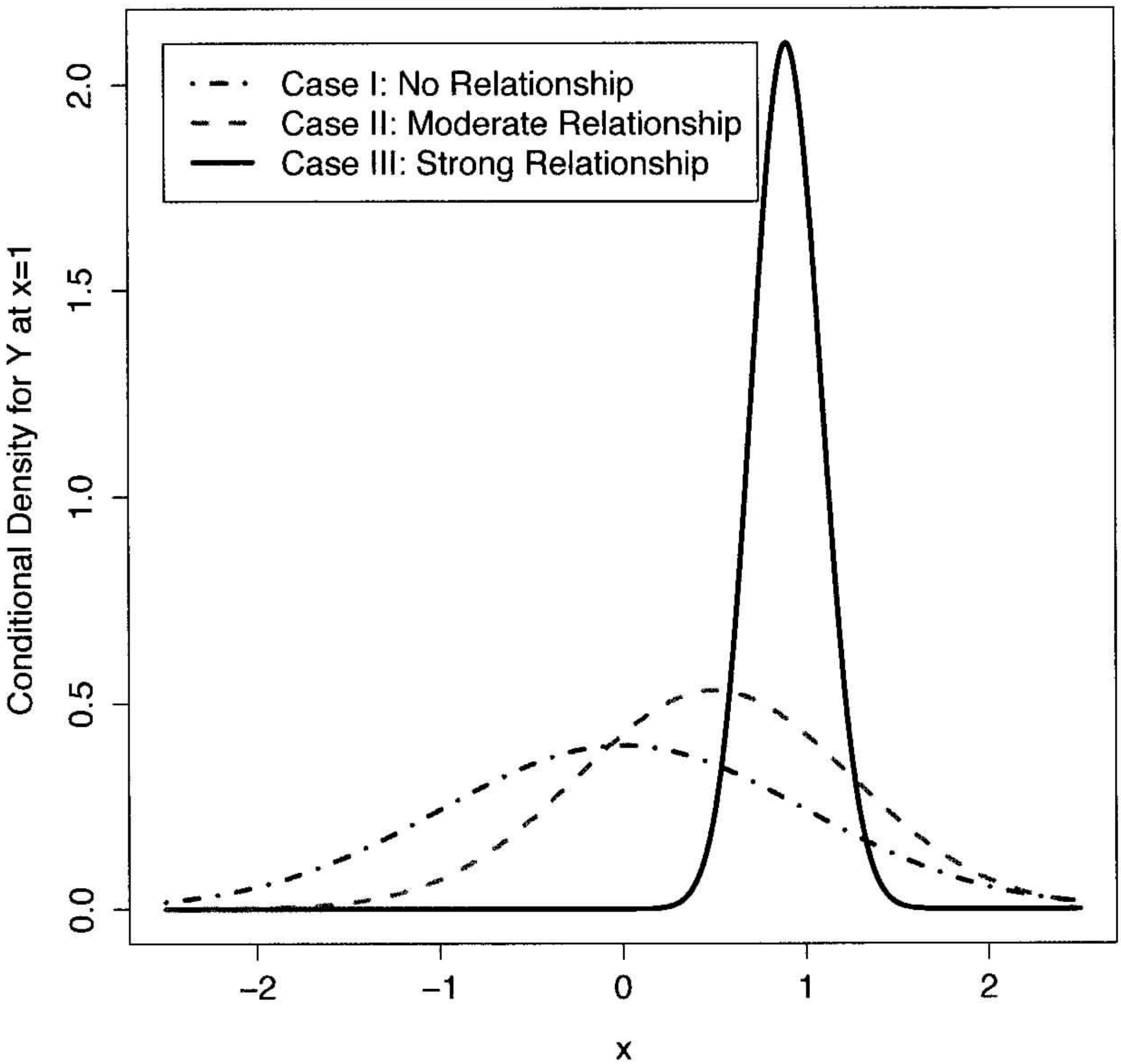
when X,Y independent,

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{f_X(x)f_Y(y)}{f_X(x)} = f_Y(y)$$

point: regardless of X conditioned on,

$$f_{Y|X}(y|x) = f_Y(y)$$

1 The conditional distribution Y for $X = 1$ changes across the three Cases.



2

Covariance

The **covariance between** X **and** Y, denoted $\text{Cov}(X, Y)$, is defined as

$$\text{Cov}(X, Y) = E[(X - E(X))\,(Y - E(Y))]$$

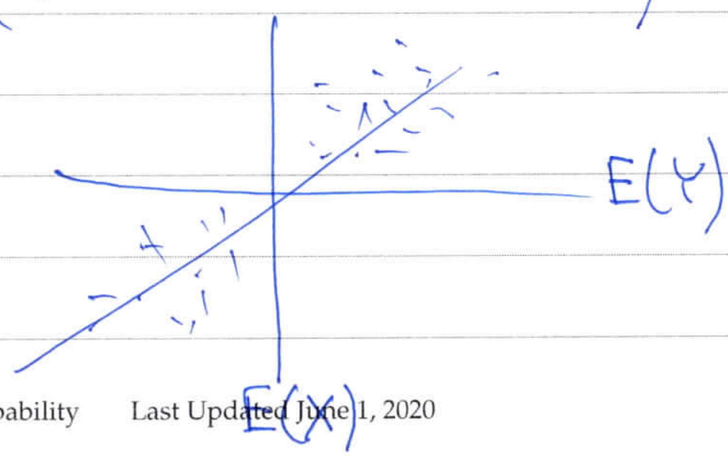
This is a measure of the amount of **linear association** between X and Y. The sign of **Cov**(X, Y) tells us the **direction** of that association.

Exercise: Looking at the definition of the covariance, explain under what conditions $\text{Cov}(X, Y) > 0$, $\text{Cov}(X, Y) < 0$, and $\text{Cov}(X, Y) = 0$.

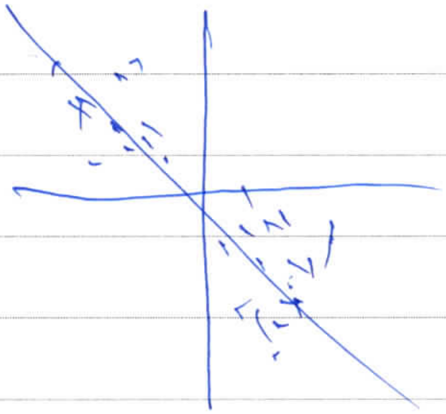
When $\text{Cov}(X,Y) > 0$?

If, when X exceeds its mean, Y also exceeds its mean, and when X is less than its mean, Y is also less than its mean, then

$$E\big((X - E(X))(Y - E(Y))\big) > 0$$



Could make a similar statement for when $Cov(X,Y) < 0$



When is $Cov(X,Y) = 0$? No linear relationship

If X, Y are independent,

$$Cov(X,Y) = E\Big((X - E(X))(Y - E(Y)) \Big)$$

$$= E\big(X - E(X)\big) E\big(Y - E(Y)\big)$$ since X, Y ind.

$$= 0$$ since $E(X - E(X)) = E(X) - E(X) = 0$

$Cov(X,Y) = 0 \Rightarrow$ "X, Y are uncorrelated"

Note: X, Y uncorrelated does <u>not</u> imply X, Y independent

Example

$Z \sim N(0,1)$, $U = \pm 1$ w.p. $1/2$

U, Z independent

$Y = UZ$, $Y(\omega) = U(\omega) Z(\omega)$

What is the dist. of Y? Y is also standard normal (derive its CDF)

$$
\begin{aligned}
Cov(Y,Z) &= E(YZ) - E(Y)E(Z) \\
&= E(YZ) \\
&= E(UZ^2) \quad \text{since } Y = UZ \\
&= E(U)E(Z^2) \quad \text{since } U, Z \text{ ind.} \\
&= 0
\end{aligned}
$$

Y, Z are uncorrelated, but Y, Z are dependent

Note that Y, Z are both normal

Instead: If (U,V) is <u>bivariate normal</u>, then U, V uncorr $\Rightarrow$ U, V ind.

When $\mathrm{Cov}(X, Y) = 0$, then X and Y are said to be **uncorrelated**.

Further, the **correlation between X and Y** is defined to be

$$\mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sqrt{V(X)V(Y)}}$$

Properties that you should know:

1. $\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$
2. $\mathrm{Cov}(X, Y) = E(XY) - E(X)E(Y)$
3. $\mathrm{Cov}(X, X) = V(X)$
4. $\mathrm{Cov}(aX + b, cY + d) = ac\,\mathrm{Cov}(X, Y)$
5. If X and Y are independent, then $\mathrm{Cov}(X, Y) = 0$.
6. $-1 \leq \mathrm{Corr}(X, Y) \leq 1$

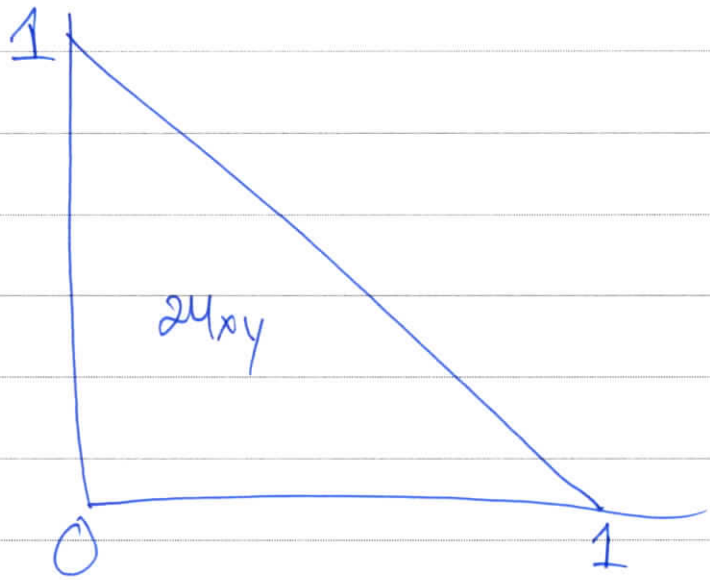
Exercise: Is there a general condition under which X and Y uncorrelated implies that X and Y are independent?

If (X, Y) is bivariate normal.

See ~~next~~ previous page

Exercise: Find $\text{Cov}(X, Y)$ for the "triangular support" example.

$(X,Y) \sim$



Recall: $X \sim \text{Beta}(2,3)$, $Y \sim \text{Beta}(2,3)$

$$E(X) = E(Y) = \frac{2}{5}$$

$$E(XY) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy\, f_{X,Y}(x,y)\,dx\,dy$$

$$= \int_{y=0}^{1}\int_{x=0}^{1-y} 24x^2y^2\,dx\,dy$$

$$= \frac{8}{60}$$

$$\text{Cov}(X,Y) = E(XY) - E(X)E(Y) = -\frac{2}{75}$$

Exercise: Suppose that X and Y are random variables with $Y = aX + b$ for constants a and b with $a \neq 0$. What is the correlation between X and Y?

$$\begin{aligned} Cov(X,Y) &= Cov(X, aX+b) \\ &= a\,Cov(X,X) \\ &= a\,V(X) \end{aligned}$$

using props. of cov.

$$Corr(X,Y) = \frac{Cov(X,Y)}{\sqrt{V(X)V(Y)}}$$

$$\boxed{V(Y) = V(aX+b) = a^2 V(X)}$$

$$= \frac{a\,V(X)}{\sqrt{a^2 V(X) V(X)}} = \frac{a}{|a|}$$

$$= sgn(a) = \begin{cases} 1, & a>0 \\ -1, & a<0 \end{cases}$$

The Bivariate Normal Distribution

The pair (X_1, X_2) has the **bivariate normal distribution** if it has joint pdf

$$f(\mathbf{x}) = \frac{1}{2\pi\sqrt{|\Sigma|}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x}-\boldsymbol{\mu})\right)$$

where $\boldsymbol{\mu}$ is the vector of means and Σ is the **covariance matrix**:

$$\Sigma = \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) \\ \mathrm{Cov}(X_1, X_2) & \mathrm{Var}(X_2) \end{bmatrix}.$$

0

0

Some important properties:

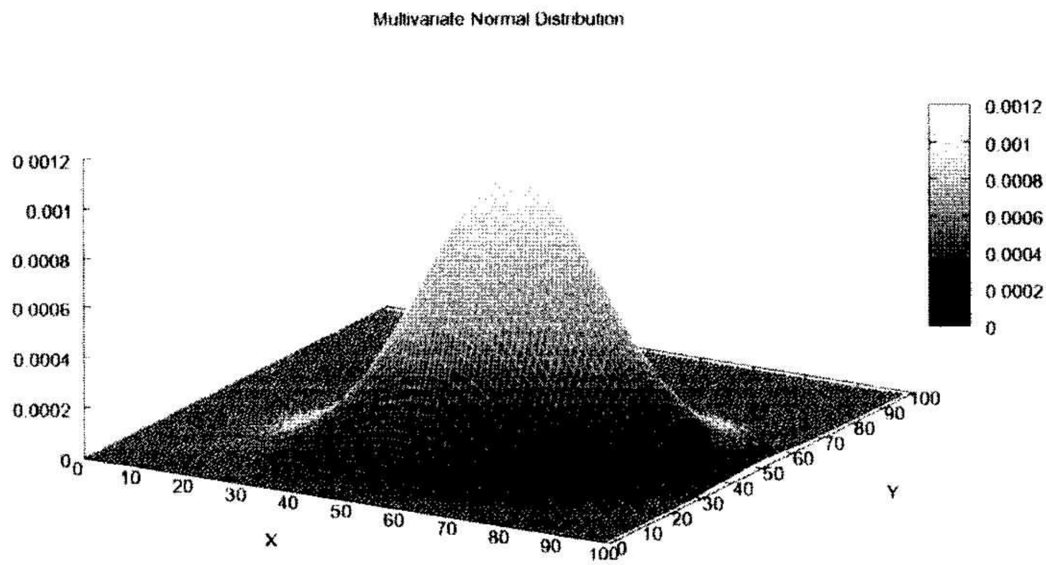
1. X_1 and X_2 are normally distributed, i.e. the marginals are normal.
2. The conditional distribution of X_1 given $X_2 = x_2$ is normal with mean
$$\mu_1 + \frac{\sigma_1}{\sigma_2}\rho(x_2 - \mu_2)$$
and variance
$$(1-\rho^2)\sigma_1^2$$
where ρ equals the correlation between X_1 and X_2.

$$\rho = \frac{\mathrm{Cov}(X_1, X_2)}{\sqrt{V(X_1)V(X_2)}}$$

3. All linear combinations of X_1 and X_2 are also normally distributed.
4. X_1 and X_2 uncorrelated implies that X_1 and X_2 are independent.

Also: ~~5.~~ X_1 and X_2 normal and independent implies that (X_1, X_2) is bivariate normal

$$\boxed{f_{\underset{\sim}{X}}(\underset{\sim}{x}) = f_{X_1}(x_1)\, f_{X_2}(x_2)}$$



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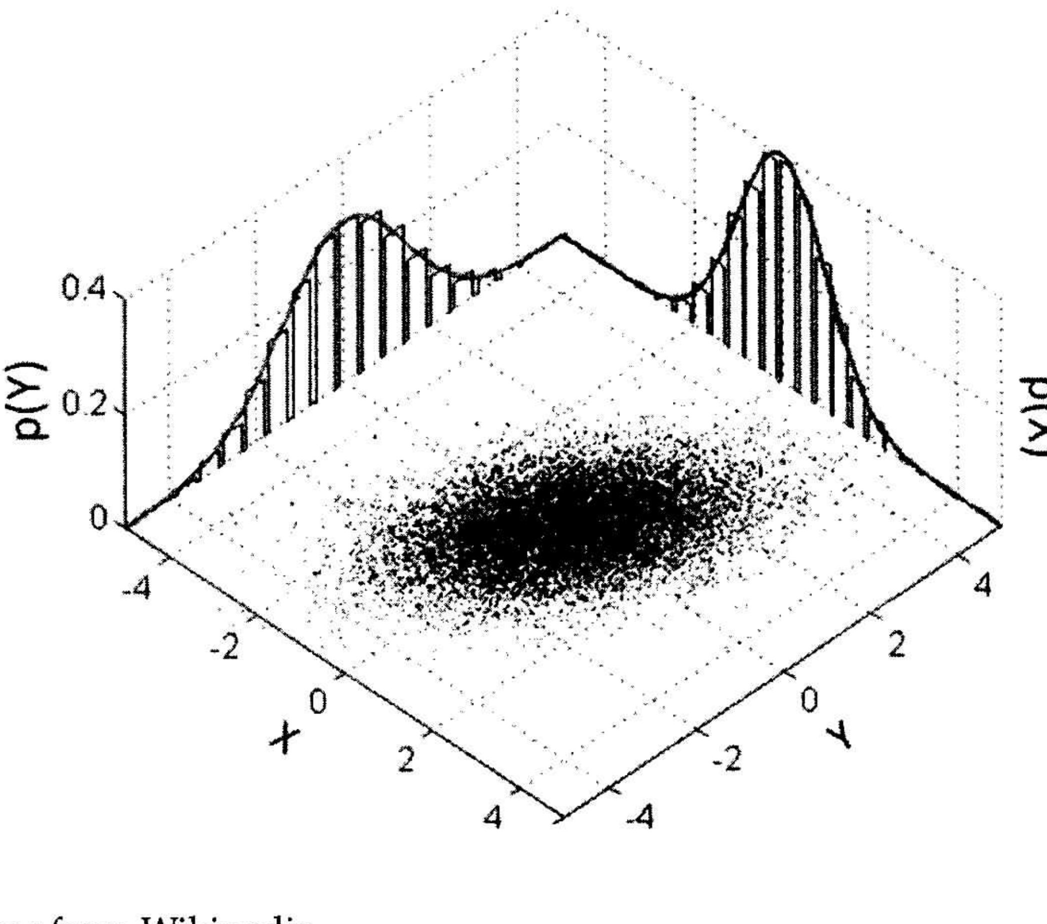

2

3 Figures from Wikipedia.

More than Two Random Variables

As one moves to the joint distribution of more than two random variables, it becomes increasingly important to work in matrix notation.

Define a **random vector** of length n by writing

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix}$$

where each of the X_i is a random variable. Then we can define

$$E(\mathbf{X}) = \begin{bmatrix} E(X_1) \\ E(X_2) \\ \vdots \\ E(X_n) \end{bmatrix}$$

and $V(\mathbf{X})$ is a n by n matrix whose (i, j) element is $\mathrm{Cov}(X_i, X_j)$. (Of course, this means that the diagonal holds the variances of the X_i.)

It is standard to let $\boldsymbol{\mu} = E(\mathbf{X})$ and $\boldsymbol{\Sigma} = V(\mathbf{X})$.

$\underset{\sim}{\mu}$ $\underset{\sim}{\Sigma}$

Some important properties: Let **u** and **v** denote two vectors of n scalars.

1. $E(\mathbf{u}^T\mathbf{X}) = \mathbf{u}^T E(\mathbf{X})$ $\qquad u^T X = \sum_{i=1}^{n} u_i X_i$
2. $V(\mathbf{u}^T\mathbf{X}) = \mathbf{u}^T \Sigma \mathbf{u}$ $\qquad V\left(\sum_{i=1}^{n} u_i X_i\right) =$
3. $\mathrm{Cov}(\mathbf{u}^T\mathbf{X}, \mathbf{v}^T\mathbf{X}) = \mathbf{u}^T \Sigma \mathbf{v}$
4. If **A** is p by n (non-random) matrix, then $E(\mathbf{AX}) = \mathbf{A}\boldsymbol{\mu}$ and $V(\mathbf{AX}) = \mathbf{A}\Sigma\mathbf{A}^T$

Exercise: Of course, the variance of a random variable cannot be negative. What is an analogous constraint that is placed on Σ?

Σ must be positive semidefinite
(i.e. $u^T \Sigma u \geq 0$ for all u)

Σ is positive definite if $u^T \Sigma u > 0$
for any $u \neq 0$

Σ p.d. $\Rightarrow$ Σ^{-1} exists

Note: If Σ is p.s.d., then
$A \Sigma A^T$ is also p.s.d.

Exercise: Suppose that X_1, X_2, and X_3 are random variables such that

$$E(X_1) = -1,\ E(X_2) = 0,\ E(X_3) = 2$$

and

$$V(X_1) = 1,\ V(X_2) = 3,\ V(X_3) = 1$$

and

$$\mathrm{Cov}(X_1, X_2) = 0.5,\ \mathrm{Cov}(X_1, X_3) = 0,\ \mathrm{Cov}(X_2, X_3) = -0.5.$$

Let

$$Y_1 = X_1 - X_2$$

and

$$Y_2 = X_1 + 2X_2 + 9X_3.$$

What are

$$E(Y_1),\ E(Y_2),\ V(Y_1),\ V(Y_2),\ \text{and } \mathrm{Cov}(Y_1, Y_2)?$$

Messy!

Exercise: Let $\mathbf{X}$ be a random vector such that

$$E(\mathbf{X}) = \boldsymbol{\mu} = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$$

and

$$V(\mathbf{X}) = \boldsymbol{\Sigma} = \begin{bmatrix} 1 & 0.5 & 0 \\ 0.5 & 3 & -0.5 \\ 0 & -0.5 & 1 \end{bmatrix}.$$

Let

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & 9 \end{bmatrix}.$$

What are the mean and variance of $\mathbf{Y} = \mathbf{AX}$?

Same question as previous page.

$$E(\underset{\sim}{Y}) = E(\underset{\sim}{A}\underset{\sim}{X}) = \underset{\sim}{A}E(\underset{\sim}{X}) = \underset{\sim}{A}\underset{\sim}{\mu} = \begin{bmatrix} -1 \\ 17 \end{bmatrix}$$

$$V(\underset{\sim}{Y}) = V(\underset{\sim}{A}\underset{\sim}{X}) = \underset{\sim}{A}V(\underset{\sim}{X})\underset{\sim}{A}^T = \begin{bmatrix} 3 & 0 \\ 0 & 78 \end{bmatrix}$$

So, $E(Y_1) = -1$, $E(Y_2) = 17$, $V(Y_1) = 3$, $V(Y_2) = 78$, $Cov(Y_1, Y_2) = 0$

Y_1, Y_2 are uncorrelated, but not sure if Y_1, Y_2 are dependent or independent

3

Joint Distributions

We can write $f_{\mathbf{X}}(\mathbf{x})$ for the **joint distribution** of $\mathbf{X}$, much as we did in the bivariate case.

$$\begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix}$$

For instance, if $\mathbf{X}$ is continuous,

$$P(X_1 \in (a_1, b_1), X_2 \in (a_2, b_2), \ldots, X_n \in (a_n, b_n)) = \int_{a_1}^{b_1} \int_{a_2}^{b_2} \cdots \int_{a_n}^{b_n} f_{\mathbf{X}}(\mathbf{x})\, dx_n\, dx_{n-1} \cdots dx_1$$

We obtain multivariate marginal and conditional distributions as before:

Integrate out to get marginals. For example,

$$f_{(X_3, X_4, \ldots, X_n)}(x_3, x_4, \ldots, x_n) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{\mathbf{X}}(\mathbf{x})\, dx_1\, dx_2$$

For the conditionals **divide by density** for the random variables being conditioned on. For example,

$$f_{(X_1, X_2) \,|\, (X_3, X_4, \ldots, X_n)}(x_1, x_2 \,|\, x_3, x_4, \ldots, x_n) = \frac{f_{\mathbf{X}}(\mathbf{x})}{f_{(X_3, X_4, \ldots, X_n)}(x_3, x_4, \ldots, x_n)}$$

Exercise: Suppose that $X_1, X_2, \ldots, X_n$ have joint pdf

$$f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} \frac{2n!}{(2+\sum_{i=1}^{n} x_i)^{n+1}}, & \text{if } x_i > 0, \ \ i = 1, 2, \ldots, n \\ 0, & \text{otherwise} \end{cases}$$

For this exercise, assume that $n = 5$.

1. What is the marginal distribution for (X_1, X_4)?

2. What is the conditional distribution of (X_2, X_3, X_5) given (X_1, X_4)?

1

$$f_{(X_1;X_4)}(x_1, x_4) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{\underset{\sim}{X}}(\underset{\sim}{x})\, dx_2\, dx_3\, dx_5$$

$$= \int_{0}^{\infty}\int_{0}^{\infty}\int_{0}^{\infty} \frac{2n!}{(2+\sum x_i)^{n+1}}, \quad \begin{matrix} x_1 > 0 \\ x_4 > 0 \end{matrix}$$

$$= \frac{4}{(2+x_1+x_4)^3}, \quad x_1 > 0, \ x_4 > 0$$

2

$$f_{(X_2,X_3,X_5 | X_1,X_4)}(x_2, x_3, x_5 | x_1, x_4)$$

$$= \frac{f_{\underset{\sim}{X}}(\underset{\sim}{x})}{f_{X_1,X_4}(x_1,x_4)} = \frac{\frac{240}{(2+\sum x_i)^6}}{\frac{4}{(2+x_1+x_4)^3}} = 60\left(\frac{(2+x_1+x}{(2+\sum x_i)^6}\right.$$

$x_i > 0$

Independence

The random variables $X_1, X_2, \ldots, X_n$ are independent if

$$f_{\mathbf{X}}(\mathbf{x}) = \prod_{i=1}^{n} f_{X_i}(x_i) \quad \text{for all } x_1, x_2, \ldots, x_n$$

When say that "$X_1, X_2, \ldots, X_n$ are i.i.d." that means that they are **independent and identically distributed**. The term "identically distributed" means that the random variables have the exact same distribution, not simply that they are "all normal" or "all exponential."

Multivariate Normal

The vector $\mathbf{X}$ (of length n) has the **multivariate normal distribution** if it has joint pdf

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^n |\boldsymbol{\Sigma}|}} \exp\left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right)$$

It is assumed that $\boldsymbol{\Sigma}$ is positive definite.

This implies that $E(\mathbf{X}) = \boldsymbol{\mu}$ and $V(\mathbf{X}) = \boldsymbol{\Sigma}$. All marginal and conditional distributions are multivariate normal. Further, any random variable/vector of the form $\mathbf{AX}$ will also be multivariate normal, provided that $\mathbf{A\Sigma A}^T$ is positive definite.

If $\mathrm{Cov}(X_i, X_j) = 0$, then X_i and X_j are independent, and if $\boldsymbol{\Sigma}$ is diagonal, then $X_1, X_2, \ldots, X_n$ are independent.

$$\Sigma = \begin{bmatrix} \sigma_1^2 & & & 0 \\ & \sigma_2^2 & & \\ & & \ddots & \\ 0 & & & \sigma_n^2 \end{bmatrix}$$